

Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

If a triangle with its vertices at $(a, 0)$ $(2a, 0)$ and (a, a) , where a is positive, has an area of 16 sq. units, what is the value of a ?

☐ a) $2\sqrt{2}$

☐ b) **4**

☐ c) $4\sqrt{2}$

☐ d) **8**

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

For $x > 0$, what is the minimum value of $\frac{(x+4)(10+x)}{x+2}$?

Q3. DIRECTIONS *for question 3:* Select the correct alternative from the given choices.

If the expressions $a_1 - 2a_2$, $2a_2 - 3a_3$, $3a_3 - 4a_4$ and $4a_4 - a_1$ are in arithmetic progression, what is the ratio of a_1 and a_2 ?

- ☐ a) **2 : 1**
- ☐ b) **4 : 1**
- ☐ c) **8 : 1**
- ☐ d) **Cannot be determined**

DIRECTIONS for questions 4 and 5: Answer the questions on the basis of the information given below.

In a six-digit number, the first, third and fifth digits, from the left, are all distinct and odd. The two-digit numbers formed by the first and second digits, the third and fourth digits and the fifth and sixth digits, from the left, taken in the order in which they appear, are all divisible by 8.

Q4. DIRECTIONS for question 4: Select the correct alternative from the given choices.

If there is only one '6' in the number, which of the following digits is necessarily there in the number?

☐ a) 1

☐ b) 5

☐ c) 7

☐ d) 9

Q5. DIRECTIONS *for question 5:* Type in your answer in the input box provided below the question.

How many of the possible numbers are divisible by 16?

Q6. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

There are two lighted candles whose rates of burning (in cm/min) are in the ratio of 2 : 1. At 6:00 p.m., their lengths are in the ratio of 3 : 2 and at 9:00 p.m., their lengths are in the ratio of 2 : 3. At what time were their lengths equal?

- ☐ a) 7:30 p.m.
- ☐ b) 8:00 p.m.
- ☐ c) 8:24 p.m.
- ☐ d) 8:36 p.m.

Q7. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

A triangle is cut out of a square, which, in turn, was originally cut out of a circle. If the diameter of the circle was 4 cm, the area of the triangle is at most

☐ a)

$4\sqrt{3}$ sq.cm.

☐ b) **4 sq.cm.**

☐ c) **8 sq.cm.**

☐ d)

$8\sqrt{2}$ sq.cm.

Q8. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

Every day on her way to college, Divya crosses a bus coming from the opposite direction, at a certain point on the way. One day Divya was late by 15 minutes and as a result she crossed the bus five minutes after the usual time. What is the ratio of the speed of the bus to that of Divya?

☐ a) **2**

☐ b) $\frac{3}{2}$

☐ c) **3**

☐ d) $\frac{5}{2}$

Q9. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

A pack of 52 playing cards comprises four suites of 13 cards each. In each suite, there are cards of 13 distinct face values – from 1 to 13. In how many ways can a pack of 52 playing cards be distributed equally among four persons so that each person receives cards of all the possible face values?

- ☐ a) ${}^{52}\mathbf{C}_{13}$
- ☐ b) $(24)^{13}$
- ☐ c) $\frac{52!}{(13!)^4}$
- ☐ d) $(13!)^4$

Q10. DIRECTIONS *for questions 6 to 10:* Select the correct alternative from the given choices.

Ajay and Bhavan are two friends who have a certain number of marbles with them. Ajay has m marbles and Bhavan has n marbles. The number m is a two-digit number whose square root is a perfect square. The number n is a two-digit prime number whose units digit and tens digit are both distinct perfect squares. Considering the following statements independently, which of the statements is/are definitely false?

I. $m < n$, if the tens digit of m is less than its units digit.

II. $|m - n|$ is a perfect square.

III. $m + n$ is a perfect square.

- ☐ a) **Only I and III**
- ☐ b) **Only II**
- ☐ c) **Only I**
- ☐ d) **None of the above**

DIRECTIONS for questions 11 and 12: Answer the questions on the basis of the information given below.

The reflection of the graph of a function $f(x)$ with respect to a line ℓ is obtained by treating the line as a mirror, being exactly halfway between each point of $f(x)$ and its reflection. An operation $R(f(x), \ell)$ is defined for the graph of any given function $f(x)$ to yield the graph of a new function $g(x)$ such that $g(x) = |h(x)|$, where $h(x)$ is the reflection of $f(x)$ with respect to the line ℓ .

Q11. DIRECTIONS for question 11: Type in your answer in the input box provided below the question.

If $f(x) = |x|$, and ℓ is $y - 1 = 0$, then find the number of points at which the graph of $R(R(f(x), \ell), \ell)$ touches the x-axis.

DIRECTIONS for questions 11 and 12: Answer the questions on the basis of the information given below.

The reflection of the graph of a function $f(x)$ with respect to a line ℓ is obtained by treating the line as a mirror, being exactly halfway between each point of $f(x)$ and its reflection. An operation $R(f(x), \ell)$ is defined for the graph of any given function $f(x)$ to yield the graph of a new function $g(x)$ such that $g(x) = |h(x)|$, where $h(x)$ is the reflection of $f(x)$ with respect to the line ℓ .

Q12. DIRECTIONS for questions 12: Select the correct alternative from the given choices.

If $f(x) = x$, ℓ_1 is $x - 1 = 0$ and ℓ_2 is $y + 1 = 0$, then which of the following is true of $R(R(f(x), \ell_1), \ell_2)$?

- ☐ a) It has a maximum value of -2
- ☐ b) It has a minimum value of -1
- ☐ c) It has a minimum value of 2
- ☐ d) It has a maximum value of 1

Q13. DIRECTIONS *for question 13:* The question below is followed by two statements, I and II. Answer the question using the following instructions:
Mark (A) if the question can be answered by any one of the two statements alone but not by the other statement alone.

Mark (B) if the question can be answered by either of the two statements alone.

Mark (C) if the question can be answered only if both the statements are taken together.

Mark (D) if the question cannot be answered even if both the statements are taken together.

If the 1st of this month was a Sunday, what day of the week was the first day of this year?

I. The 1st of the next month is a Wednesday.

II. The 1st of the previous month was a Saturday.

☐ a) **A**

☐ b) **B**

☐ c) **C**

☐ d) **D**

Q14. DIRECTIONS *for question 14:* Type in your answer in the input box provided below the question.

Consider a natural number m , ($m > 2$), and $n = 2m + 1$. Let X be the sum of the two-digit numbers as shown below (each two-digit number written in the number system to the base n):

$$X = (12)_n + (34)_n + \dots + ([2m - 1] [2m])_n$$

Express $X + m + 2$ in base m .

Q15. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

If $\log(x + 6) = \log x + \log 6$, find the value of x .

- ☐ a) 1
- ☐ b) 1.2
- ☐ c) 1.5
- ☐ d) $\frac{\log 6}{\log x}$

Q16. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

Find the product of the roots of the equation $(2x^2 + 13)(x - 3) + (2x^2 + 13)(x - 5) = 0$.

☒ a) 26 ✓ **Your answer is correct**

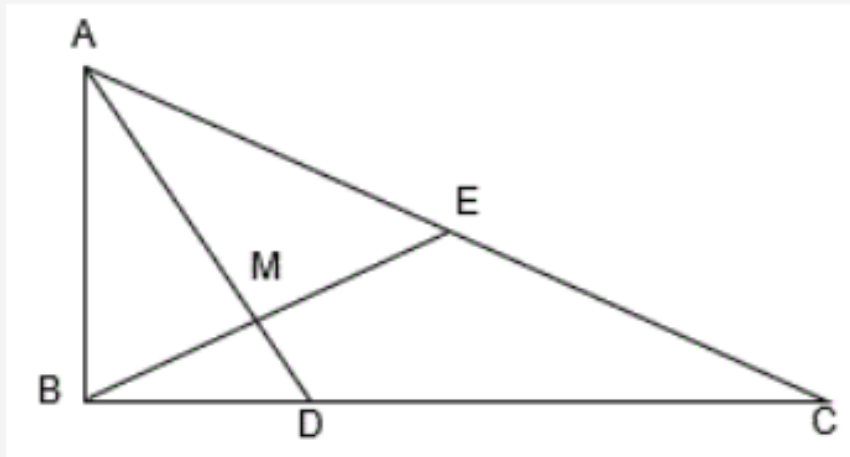
☐ b) -26

☐ c) 39

☐ d) -39

Q17. DIRECTIONS for questions 15 to 18: Select the correct alternative from the given choices.

In the figure below, if $BD : DC = 1 : 3$ and $AE : EC = 1 : 1$, find $AM : MD$.



- ☐ a) **3 : 2**
- ☐ b) **3 : 1**
- ☐ c) **2 : 1**
- ☐ d) **4 : 1**

Q18. DIRECTIONS *for questions 15 to 18:* Select the correct alternative from the given choices.

A company introduces a scheme to control the rate of work (i.e., work done per unit time) of its employees. According to this scheme, a standard rate of work is specified. If an employee works at a lower rate than this, he would receive 8% of the standard rate of pay (in ₹/hr) less for every 5% (of the standard rate of work) drop in his rate of work. Ashok, an employee of that company completed a job in exactly three hours. Given that he would have taken 25% less time had he worked at the standard rate of work, and the standard rate of pay is ₹6/hr, how much was he actually paid for the job?

- ☐ a) ₹8.40
- ☐ b) ₹12.24
- ☐ c) ₹10.80
- ☐ d) ₹9.00

Q19. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

Mr. Ramnath took an interest-free loan of ₹2,00,000 for constructing a house, with the condition that he would repay the entire loan in monthly instalments. The first instalment would be of ₹2,000 and every subsequent instalment would increase by ₹500. What is the value (in ₹) of the last instalment that Mr. Ramnath paid?

Q20. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

There are three cities A, B and C. Each of these cities is connected with the other two cities with at least one direct road. We can travel from one city to another city either by a direct road connecting the two cities or by travelling through the third city. If the total number of routes from A to B and from B to C are 25 and 17 respectively, find the number of direct roads from A to C.

Q21. DIRECTIONS *for question 21:* Select the correct alternative from the given choices.

Three persons – Amar, Akbar and Anthony, working together, complete a piece of work in a certain number of days. The number of days they take is 20 less than what Amar alone would have taken; 8 less than what Akbar alone would have taken; and half of what Anthony alone would have taken. How many days do they take to complete the work, when working together?

- ☐ a) 2
- ☒ b) 4 ✓ **Your answer is correct**
- ☐ c) 6
- ☐ d) 8

Q22. DIRECTIONS *for question 22:* Type in your answer in the input box provided below the question.

If $f(x) = \frac{25^x}{25^x + 5}$, find the value of $f\left(\frac{1}{99}\right) + f\left(\frac{2}{99}\right) + \dots + f\left(\frac{98}{99}\right)$.